

Walmart Sourcing & Procurement Intelligence Platform

Comprehensive Technical Project Summary, RAG Architecture & System Blueprint

Project Scope: Walmart Category Management RAG Engine	Parsing Engine: IBM Docling (DocumentConverter)
Primary Vector Store: PostgreSQL 16 + pgvector	LLM Backend: Google Gemini 2.5 Flash / Flash Lite
Embedding Model: text-embedding-004 (768d)	Deployment: Docker Compose Multi-Container

1. Executive Summary & Core Business Objectives

The **Walmart Sourcing & Procurement Intelligence Platform** is an enterprise-grade Retrieval-Augmented Generation (RAG) system engineered specifically for Walmart category buyers, procurement managers, and legal reviewers. When evaluating complex vendor proposals, service level agreements (SLAs), and master supply agreements (MSAs), sourcing teams typically spend days manually comparing incoming bids against Walmart's master procurement baselines and historical category precedents.

This platform automates contract comparison, vendor proposal gap analysis, precedent benchmarking, and interactive negotiation battlecard generation—reducing review cycles from days to under 60 seconds while providing 100% grounded citations to raw contract clauses.

2. Primary Sourcing & Procurement Use Cases

- **Vendor Proposal Evaluation & Gap Detection:** Automated side-by-side contract comparison highlighting explicit deviations, un-capped liabilities, payment term discrepancies (e.g. Net 90 vs Net 30), and omitted SLA metrics relative to Walmart baseline expectations.
- **Precedent Knowledgebase Benchmarking:** Indexing historical supplier agreements across product categories (Activewear, Footwear, Electronics, Logistics) to empower sourcing managers with past precedent benchmarks during negotiation.
- **Priority Negotiation Battlecard Generation:** Extracting high-risk contract deviations and ranking them by priority with precise clause citations, summary of variance, and exact contract source references.
- **Category-Aware Interactive AI Assistant:** Context-grounded chatbot allowing procurement officers to query vendor commitments, liability caps, audit rights, and termination clauses across indexed documents.

3. Platform Architecture & System Workflow

The system follows a modern microservice architecture structured into four decoupled layers: Document Parsing, Vector Indexing, LLM Synthesis, and Tiered Frontend Visualization.

Layer	Technology / Tool	Role & Operation
1. Document Parsing	IBM Docling (DocumentConverter)	Converts PDF, DOCX, HTML into layout-aware Markdown trees while maintaining headers, tables, and page boundaries.
2. Chunking Engine	Layout-Aware Splitter	Splits text into <i>Story</i> (narrative context) and <i>Criteria</i> (clause-level granularity) chunks with rich JSON metadata.

3. Embedding & Vector DB	Google Gemini Embedding + PostgreSQL pgvector	Generates 768-dim embeddings (`text-embedding-004`) stored with IVFFlat cosine similarity indexes.
4. Gap Analysis LLM	Google Gemini 2.5 Flash (Temperature = 0.2)	Executes grounded JSON gap analysis comparing incoming proposals against baseline SLAs with Pydantic validation.
5. UI Dashboard	React + TypeScript + Tailwind CSS	Renders Tier 1 Sub-scores, Tier 2 Negotiation Battlecard, and Tier 3 Collapsible Section Accordions.

4. Frontend UI Workflow & Tiered Battlecard Design

The frontend user interface is organized into three distinct analysis tiers to serve both executive overview needs and granular legal inspection:

• **Tier 1: High-Level Executive Summary & Scorecard**

Displays an overall contract compatibility score (0-100%) calculated across weighted domain criteria. Includes visual status count pills highlighting **Compliant**, **Deviation**, and **Gap** metrics at a glance.

• **Tier 2: Priority Negotiation Battlecard**

Streamlined for maximum clarity during vendor calls. Unnecessary leverage and counter-position boxes have been completely removed. Each battlecard item features: **Priority Tag (#1, #2...)**, **Clause Category Tag**, **Concise Deviation Summary**, and an interactive collapsible drawer: **View Citation & Precedent Benchmark**.

• **Tier 3: Section-Wise Analysis (Tier 3 Section-Wise Analysis)**

Groups contract clauses by section (e.g. Legal & Liability, Payment & Commercials, Logistics & Delivery). Section accordions default to collapsed (`open = false`) and expand upon click to reveal clause-by-clause baseline vs proposal comparisons.

5. Database Schema & Inspection Commands

The system utilizes PostgreSQL 16 with the `pgvector` extension. The central relation is `document\_chunks`:

Column Name	Data Type	Description
`chunk_id`	UUID (Primary Key)	Unique identifier for each document chunk.
`chunk_type`	VARCHAR(20)	Identifies chunk granularity (`story` or `criteria`).
`content`	TEXT	Extracted clause or section text content.
`embedding`	vector(768)	Gemini 768-dimensional dense vector with IVFFlat index.
`story_id`	UUID	Parent document / story relation ID.
`metadata`	JSONB	Contains page numbers, section headers, document title, category tag.
`source_path`	TEXT	Original file path or upload filename.

Live Database Inspection Command:

```
docker exec -it rag-db-1 psql -U postgres -d rag_db -c "SELECT chunk_id, chunk_type, source_path FROM document_chunks LIMIT 5;"
```

6. Tool Selection Trade-Off Analysis

Component	Selected Tool	Alternatives	Selection Rationale
Document Parsing	IBM Docling	PyPDF, Unstructured	Native layout awareness; preserves markdown table structures & section hierarchy perfectly for legal contracts.
Vector Store	PostgreSQL + pgvector	Pinecone, ChromaDB	Eliminates external SaaS dependency; unifies relational metadata (JSONB) with vector search in a single ACID DB.
LLM Engine	Google Gemini 2.5 Flash	OpenAI GPT-4o, Ollama	1M token context window allows full document pair injection; exceptional structured JSON enforcement speed.
Deployment	Docker Compose	Bare-metal, Kubernetes	Guarantees reproducible multi-container environment across developer laptops and production servers.

7. Anti-Hallucination & Reliability Guardrails

To ensure zero hallucination in high-stakes contract reviews, 6 strict safety mechanisms are enforced:

- **1. Layout Hierarchy Preservation:** Docling extracts exact section titles and paragraph boundaries, eliminating artificial text split errors.
- **2. Dual-Anchor Citation Mandate:** LLM system prompts explicitly require citing exact document names and section titles for every gap identified.
- **3. Direct Raw Context Window Injection:** Both incoming proposal text and baseline contract text are passed simultaneously in the context window.
- **4. Low Temperature Decoding:** Extraction and gap analysis run at T=0.2 to enforce deterministic, factual output generation.
- **5. Strict Pydantic JSON Schema Validation:** All LLM outputs are parsed into strict Pydantic data models with mandatory field checks.
- **6. Cosine Distance Traceability:** Vector search results include distance metrics (1.0 - distance) ensuring full traceability back to source DB chunks.

8. Core System Prompts Overview

The platform relies on three specialized system prompts located in `src/rag_ingest/prompts/`:

- ``GAP_ANALYSIS_PROMPT``: Enforces side-by-side contract comparison against Walmart SLA baselines, returning Tier 1 score breakdown, Tier 2 priority battlecard items, and Tier 3 section accordions.
- ``INGESTION_PROMPT``: Classifies incoming Markdown documents into structured story vs criteria chunks, assigning category metadata and clause tags.
- ``CHAT_PROMPT``: Governs the sourcing chatbot, ensuring all answers are grounded strictly in retrieved pgvector context with explicit file citations.

9. Docker Containerization & Environment Setup

The system is containerized into three primary services defined in `docker-compose.yml`:

- ``rag-db-1``: PostgreSQL 16 with ``pgvector/pgvector:pg16`` image listening on host port 5433 (container port 5432).
- ``rag-backend-1``: FastAPI Python 3.11 service listening on port 8000.

- **rag-frontend-1**: React Vite application served via Nginx on port 3000.

Key Environment Variables (.env):

```
GEMINI_API_KEY=AIzaSy...  
POSTGRES_USER=postgres  
POSTGRES_PASSWORD=postgres  
POSTGRES_DB=rag_db  
POSTGRES_HOST=db  
POSTGRES_PORT=5432  
LLM_MODEL=gemini-2.5-flash
```

10. Future Platform Enhancements & Enterprise Roadmap

- **ERP / SAP Ariba Integration:** Direct API integration with SAP Ariba and Coupa for automated contract ingestion upon supplier upload.
- **Automated DOCX Redlining:** Exporting Tier 2 negotiation battlecards directly into tracked-changes Word documents (.docx) for vendor sharing.
- **Multi-Modal Proposal Parsing:** Utilizing Gemini Vision capabilities to extract complex tables, workflow diagrams, and price schedules embedded as images.